Chapter 12

Electrodynamics and Relativity

Problem 12.1

Problem 12.2

Problem 12.3

Problem 12.4

As the outlaws escape in their getaway car, which goes $\frac{3}{4}c$, the police officer fires a bullet from the pursuit car, which only goes $\frac{1}{2}c$. The muzzle velocity of the bullet (relative to the gun) is $\frac{1}{3}c$. Does the bullet reach its target (a) according to Galileo, (b) according to Einstein?

Solution:

(a) Adding up the velocities with Galileo's velocity addition rule, the resulting velocity of the bullet is

$$\frac{1}{2}c + \frac{1}{3}c = \frac{5}{6}c > \frac{3}{4}c$$

so the bullet does reach its target.

(b) Using Einstein's velocity addition rule, the result is instead

$$v_{\text{bullet}} = \frac{v_{\text{car}} + v_{\text{muzzle}}}{1 + \frac{v_{\text{car}} v_{\text{muzzle}}}{c^2}} = \frac{\frac{5}{6}c}{1 + \frac{1}{6}} = \frac{5}{7}c < \frac{3}{4}c$$

so the bullet does not reach its target. □

Problem 12.5

Problem 12.6

Problem 12.7

In a laboratory experiment, a muon is observed to travel 800 m before disintegrating. A graduate student looks up the lifetime of a muon (2×10^{-6}s) and concludes that its speed was

$$v = \frac{800 \text{ m}}{2 \times 10^{-6}\text{s}} = 4 \times 10^{8} \text{ m/s}.$$

Faster than light! Identify the student's error, and find the actual speed of this muon.

Solution:

The student did not take into account the lifetime of the muon in the muons reference time. The muons lifetime in its own reference time is

$$\gamma t = \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} t$$

If it travels for 800m then its speed is

$$\frac{800 \text{ m}}{\gamma \times 2 \times 10^{-6}\text{s}} = v$$

Simplifying the distance over time gives

$$\frac{800 \text{ m}}{2 \times 10^{-6}\text{s}} = \frac{4}{3} c$$

Solving for v

$$\frac{4}{3}c = \gamma v = \frac{v}{\sqrt{1-\left(\frac{v}{c}\right)^2}}$$
$$\downarrow$$
$$\left(\frac{4}{3}c\right)^2 = \frac{v^2}{1-\left(\frac{v}{c}\right)^2}$$
$$\downarrow$$
$$\left(\frac{4}{3}c\right)^2\left(1-\left(\frac{v}{c}\right)^2\right) = v^2$$
$$\downarrow$$
$$\left(\frac{4}{3}c\right)^2 = v^2\left(1+\left(\frac{4}{3}c\right)^2\frac{1}{c^2}\right) = v^2\left(1+\frac{16}{9}\right)$$
$$\downarrow$$
$$v = \frac{4}{3}c\frac{1}{\sqrt{1+\frac{16}{9}}} = \frac{4}{3}c\frac{1}{\sqrt{\frac{25}{9}}} = \frac{12}{15}c = \boxed{\frac{4}{5}c}$$

□

Problem 12.8

Problem 12.9

A Lincoln Continental is twice as long as a VW Beetle, when they are at rest. As the Continental overtakes the VW, going through a speed trap, a (stationary) policeman observes that they both have the same length. The VW is going at half the speed of light. How fast is the Lincoln going?

Solution:

The length of the cars while stationary is denoted L. With length contraction due to the velocity accounted for the length is denoted L'. With v_{vw} given and $L_{\text{lc}} = 2L_{\text{vw}}$, the solution is obtained by setting $L'_{\text{lc}} = L'_{\text{vw}}$ and solving for v_{lc}.

$$L_{\text{lc}} = 2L_{\text{vw}}$$

$$L'_{\text{vw}} = L_{\text{vw}}\sqrt{1-\frac{v_{\text{vw}}^2}{c^2}} = L_{\text{vw}}\frac{\sqrt{3}}{2}$$

$$L'_{\text{vw}} = L'_{\text{lc}} = L_{\text{lc}}\sqrt{1-\frac{v_{\text{lc}}^2}{c^2}} = L_{\text{vw}}\frac{\sqrt{3}}{2}$$

Using $L_{\text{lc}} = 2L_{\text{vw}}$ and solving for v_{lc}, the velocity for which the length of the Lincoln Continental equals the length of the VW Beetle is

$$v_{\text{lc}} = \frac{\sqrt{13}}{4}c$$

□

Problem 12.10

Problem 12.11

Problem 12.12

Solve

$$\text{(i) } \overline{x} = \gamma(x - vt)$$
$$\text{(ii) } \overline{y} = y$$
$$\text{(iii) } \overline{z} = z$$
$$\text{(iv) } \overline{t} = \gamma\left(t - \frac{v}{c^2}x\right)$$

for x, y, z, t in terms of $\overline{x}, \overline{y}, \overline{z}, \overline{t}$, and check that you recover

$$\text{(i') } x = \gamma(\overline{x} + v\overline{t})$$
$$\text{(ii') } y = \overline{y}$$
$$\text{(iii') } z = \overline{z}$$
$$\text{(iv') } t = \gamma\left(\overline{t} + \frac{v}{c^2}\overline{x}\right)$$

Solution:

From (i) and (iv) we have

$$\text{(i)}\quad \overline{x} = \gamma(x - vt) \to x = \frac{\overline{x}}{\gamma} + vt$$

$$\text{(iv)}\ \bar{t} = \gamma\left(t - \frac{v}{c^2}x\right) \to t = \frac{\bar{t}}{\gamma} + \frac{v}{c^2}x$$

Inserting t from (iv) into the expression for x in (i) gives

$$x = \frac{\overline{x}}{\gamma} + v\left(\frac{\bar{t}}{\gamma} + \frac{v}{c^2}x\right) \to x\left(1 - \frac{v^2}{c^2}\right) = \frac{\overline{x}}{\gamma} + \frac{v\bar{t}}{\gamma}$$

$$\downarrow$$

$$x\frac{1}{\gamma^2} = \frac{1}{\gamma}(\overline{x} + v\bar{t}) \to \boxed{x = \gamma(\overline{x} + v\bar{t}) = \text{(i')}}$$

Putting this expression for x into the expression for t in (iv) gives

$$t = \frac{\bar{t}}{\gamma} + \frac{v}{c^2}(\gamma(\overline{x} + v\bar{t})) = \gamma\left(\frac{\bar{t}}{\gamma^2} + \frac{v}{c^2}\overline{x} + \frac{v^2}{c^2}\bar{t}\right) = \gamma\left(\left(1 - \frac{v^2}{c^2} + \frac{v^2}{c^2}\right)\bar{t} + \frac{v}{c^2}\overline{x}\right)$$

$$\downarrow$$

$$\boxed{t = \gamma\left(\bar{t} + \frac{v}{c^2}\overline{x}\right) = \text{(iv')}}$$

(ii) $\longleftrightarrow$ (ii') and (iii) $\longleftrightarrow$ (iii') follow directly from the equations.

□

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